CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 21: Solids

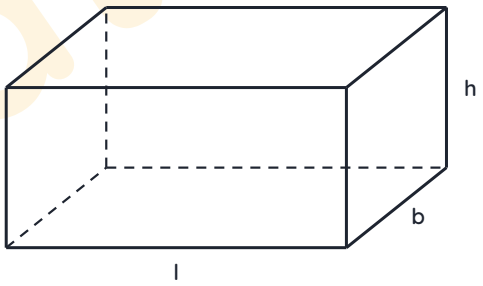
Topic-wise short notes with solved examples

Surface area and volume of cuboids and cubes, solids with a uniform cross-section, and problems on capacity, flow of water and melting and recasting.

1 Cuboid

- A **cuboid** has 6 rectangular faces, 12 edges and 8 vertices. Its dimensions are length l, breadth b and height h.

Quantity	Formula
Volume	$l \times b \times h$
Total surface area	$2(lb + bh + hl)$
Lateral surface area (four walls)	$2(l + b)h$
Diagonal	$\sqrt{l^2 + b^2 + h^2}$
Sum of all the edges	$4(l + b + h)$



Cuboid with length l, breadth b and height h

- The area of the four walls of a room is the lateral surface area.
- An **open box** has no top: surface area $= lb + 2(l + b)h$.

Example 1.1 A cuboid is 12 cm long, 9 cm broad and 8 cm high. Find its volume, total surface area and diagonal.

SOLUTION

Volume $= 12 \times 9 \times 8$. Surface area $= 2(108 + 72 + 96)$. Diagonal $= \sqrt{144 + 81 + 64}$.

864 cu cm; 552 sq cm; 17 cm

Example 1.2 A room is 8 m long, 6 m wide and 4 m high. Find the cost of painting its four walls and ceiling at ₹50 per sq m.

SOLUTION

Walls: $2(8+6)\times 4 = 112$ sq m. Ceiling: 48 sq m. Total 160 sq m.

₹8000

2 Cube

- A **cube** is a cuboid with all edges equal to a.

Quantity	Formula
Volume	a^3
Total surface area	$6a^2$
Lateral surface area	$4a^2$
Diagonal	$a\sqrt{3}$

Example 2.1 The total surface area of a cube is 294 sq cm. Find its edge, volume and diagonal.

SOLUTION

$6a^2 = 294$, so $a = 7$ cm.

7 cm; 343 cu cm; $7\sqrt{3}$ cm

Example 2.2 Three cubes of edges 3 cm, 4 cm and 5 cm are melted to form one cube. Find its edge.

SOLUTION

Volume $= 27 + 64 + 125 = 216$ cu cm

6 cm

Example 2.3 How many cubes of edge 4 cm can be cut from a cuboid 16 cm by 12 cm by 8 cm?

SOLUTION

$$\frac{16\times 12\times 8}{4\times 4\times 4}$$

24

3 Units and capacity

Conversion	Value
1 cu m	1000 litres $= 1{,}000{,}000$ cu cm
1 litre	1000 cu cm
1 cu cm	1 mL
1 kilolitre	1 cu m

- Mass $=$ volume $\times$ density.

Example 3.1 A tank is 3 m long, 2 m wide and 1.5 m deep. How many litres of water does it hold?

SOLUTION

Volume $= 9$ cu m

9000 litres

4 Hollow boxes

- For a box made of material of thickness t (closed on all sides): internal dimensions $=$ external dimensions $-2t$.
- For an **open** box, the internal height is the external height $-t$.
- Volume of material $=$ external volume $-$ internal volume.

Example 4.1 A closed wooden box has external dimensions 50 cm, 40 cm and 30 cm, and the wood is 2 cm thick. Find the volume of the wood.

SOLUTION

Internal dimensions: 46 cm, 36 cm, 26 cm.

$60000 - 43056$

16,944 cu cm

5 Uniform cross-section

- A solid whose cross-section is the same all along its length (a prism) has

$$\text{Volume} = \text{area of cross-section} \times \text{length}$$

$$\text{Lateral surface area} = \text{perimeter of cross-section} \times \text{length}$$

- Total surface area $=$ lateral surface area $+2\times$ area of cross-section.

Example 5.1 A swimming pool is 40 m long and 15 m wide. Its depth increases uniformly from 1 m at one end to 3 m at the other. Find the volume of water it holds.

SOLUTION

The cross-section is a trapezium with parallel sides 1 m and 3 m and width 40 m: area $= \frac{1}{2}(1 + 3) \times 40 = 80$ sq m.

Volume $= 80 \times 15$

1200 cu m

Example 5.2 A metal bar 2 m long has a cross-section that is a right triangle with legs 6 cm and 8 cm. Find its volume and lateral surface area.

SOLUTION

Area of cross-section $= 24$ sq cm; hypotenuse 10 cm; perimeter 24 cm; length 200 cm.

4800 cu cm; 4800 sq cm

6 Flow of water

- Volume of water flowing in time t = area of cross-section of the pipe or canal $\times$ speed $\times$ time.
- Keep all lengths in the same unit.

Example 6.1 Water flows at 5 m/s through a rectangular channel 2 m wide and 1.5 m deep. How many litres flow in one minute?

SOLUTION

$2 \times 1.5 \times 5 \times 60 = 900$ cu m

9,00,000 litres

Example 6.2 A river 3 m deep and 40 m wide flows at 2 km/h. How much water falls into the sea in one minute?

SOLUTION

Speed $= \frac{2000}{60}$ m per minute. Volume $= 3 \times 40 \times \frac{2000}{60}$.

4000 cu m

Example 6.3 A field 500 m by 200 m is covered by rain to a depth of 2 cm. Find the volume of water.

SOLUTION

$500 \times 200 \times 0.02$

2000 cu m

7 Common mistakes

- Forgetting to subtract twice the thickness for a closed box.
- Using $2(lb + bh + hl)$ when only the walls or an open box are meant.
- Mixing centimetres and metres.
- Writing 1 cu m $=$ 100 litres. It is 1000 litres.
- For flow problems, forgetting to multiply by the time.
- Thinking that doubling the edge of a cube doubles the volume. The volume becomes 8 times and the surface area 4 times.

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